

Optimizing Delays in Random Access Procedure of DECT-2020 by Estimation of Device States

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Abstract—Massive Machine-Type Communication (mMTC) enables the seamless connectivity of billions energy-efficient devices, forming the intelligent infrastructure for smart cities, automated industries, and real-time data-driven decision making. In this paper we investigate the modern mMTC technology Digital Enhanced Cordless Telecommunication New Radio (DECT-2020 NR), focusing on random access (RA) procedure. The study develops a mathematical model characterizing the RA phase and proposes a hidden Markov model (HMM)-based method for estimating the number of competing devices. The research introduces and evaluates a delay reduction strategy wherein the RA phase duration is adapted based on the HMM-based estimation of the device state. Numerical results demonstrate the effectiveness of the proposed approach in optimizing RA performance. The findings contribute to improving DECT-2020 network efficiency by balancing RA and data transmission phases while minimizing latency.

Index Terms—mMTC, DECT-2020, random access, delay, hidden Markov model

I. INTRODUCTION

The Digital Enhanced Cordless Telecommunication New Radio (DECT-2020 NR) technology has emerged as a pivotal enabler for next-generation massive Machine-Type Communication (mMTC) networks [1], which are fundamental to the autonomous operation of smart cities and industrial systems [2]. Its relevance stems from its ability to support low-latency automated control and high-density device connectivity, owing to its energy-efficient and decentralized architecture.

DECT-2020 facilitates a wide range of communication scenarios, operating with various devices. These devices can be sensors that transmit data sporadically, with stringent energy constraints, or end devices or relays that handle substantial data volumes, either intermittently or continuously. This diversity necessitates the coexistence of two phases: a random access (RA) phase, which caters to energy-limited devices,

and a scheduled data transmission (DT) phase, serving devices with higher throughput demands.

On the one hand, the dual-phase structure of DECT-2020 is similar to that of Cellular IoT (CIoT) standards such as NB-IoT and LTE-M [3], [4]. However, the RA procedure used in cellular networks, which includes a set of orthogonal preambles, is significantly different from the DECT-2020 RA procedure, which is based on traditional binary exponential backoff [5]. On the other hand, well-studied systems that rely solely on binary exponential backoff [6] are not suitable for analyzing the DECT-2020 RA, which coexists with DT. Specifically, devices in DECT-2020 do not immediately initiate backoff intervals after data generation or channel sensing, but instead synchronize at the beginning of the RA phase and execute backoff procedures simultaneously based on the number of collisions they experience.

The latest IEEE 802.11 networks that implement RA Window (RAW) and Traffic Indication Map (TIM) mechanisms appear to be the most similar to DECT-2020 technology, as they use a predefined portion of resources for RA for a group of stations, which start RA synchronously at the same time. However, no completely relevant model has been found, as some studies suggest that traffic essentially passes RA during its first RAW, or is dropped [7]. Additionally, most studies focus on energy consumption and throughput, rather than delays [7]–[9]. Furthermore, even when considering RAW slots, which are similar to the RA phase in DECT-2020, the authors use approximations for transmission probability [8], [9], which can be avoided. Therefore, building models for DECT-2020 networks remains an ongoing challenge.

Previous studies have demonstrated that there is an optimal allocation of time resources for the RA phase providing minimum latency and throughput [3], [8]. However, this optimum can only be found with a known number of devices competing for the medium in the RA, the arrival intensity, and the probability of transmission attempts. For networks operating

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solely under RA, several estimation techniques, including Kalman filtering [10] and hidden Markov models (HMMs) [11], have been proposed to optimize backoff parameters. However, in hybrid systems that combine both RA and DT, estimating the number of devices in the RA phase remains an open research problem. Addressing this gap is crucial for optimizing the duration of RA phase and reducing packet delays.

In this work, we:

- Introduce a mathematical model characterizing the RA phase in DECT-2020 network.
- Propose a HMM-based method for estimating the number of competing devices in the RA.
- Develop a delay reduction strategy for the RA phase based on observation of device states, and evaluate method's effectiveness.

The remainder of this paper is organized as follows. Section II details the system model, including the specifics of the DECT-2020 RA procedure and the formalization of our Markovian model. Section III introduces the HMM-based estimation framework. Section IV presents and discusses the numerical results, and Section V concludes the paper.

II. SYSTEM MODEL

A. DECT-2020 Specifications

The DECT-2020 NR network architecture operates on a cluster mesh topology, with each end device, called a Portable Termination Point (PT), attached to one of Fixed Termination Points (FTs) devices that coordinate local radio resources. These devices forward acquired data to the backend either directly or through another FT. Every FT with associated PTs forms a network cluster. Additionally, each PT may become a FT and create its own network cluster, forming a decentralized mesh network. Once a PT becomes a FT, it begins periodically transmitting a Cluster Beacon Message (CBM), identifying the managed local radio resource parameters [12]. We refer to the time between two CBMs as the beacon interval.

A key feature of the Medium Access Control (MAC) layer in DECT-2020 NR is the coexistence of random and scheduled access [13], which can be implemented as two separate phases: RA and DT. The RA mechanism is based on binary exponential backoff, enabled by Listen Before Talk (LBT), with parameters $CW_{\min}$, defining the minimum contention window size, and N , defining the duration of RA in time slots. The last RA slot is reserved for PT to send a response message to the cluster's FT. Upon Protocol Data Unit (PDU) generation, the PT listens to the CBM transmitted by the cluster's FTs to find the beginning of the RA phase and its duration N . At the beginning of RA, the PT that has attempted to transmit the current packet b times, randomly selects a backoff period between 1 and $\min(N - 1, 2^b CW_{\min})$. We define $b_{\max}$ as

$$b_{\max} = \min(b : 2^b CW_{\min} \geq N - 1, \quad b \in \mathbb{N}). \quad (1)$$

Upon expiring backoff, the PT senses the channel and, if the channel is free, it attempts transmission. Otherwise, it stays still and waits for the channel to become free.

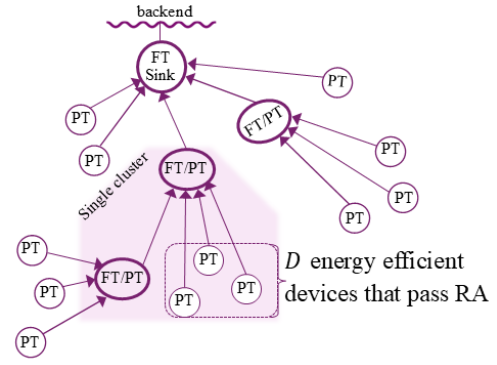


Fig. 1: An illustration of the DECT-2020 cluster that will be considered.

B. Deployment Scenario

We consider a single cluster of a multi-hop DECT-2020 network that consists of a single FT device and several PTs that wish to transmit their data to the backend via FT. We assume that the number of devices within the cluster is fixed, with D energy-efficient devices among the PTs and the remaining PTs being FT/PT devices for other clusters. The energy-efficient devices go to sleep when they are not transmitting data and therefore must pass RA before each transmission, while the FT/PTs that carry data from previous hops are always synchronized with the network and do not use the RA phase to send data through DT. An illustration of the considered DECT-2020 network part is given in Fig. 1. We assume that each PT that is currently idle (i.e., has no packet to send) generates a new packet in the interval between two RA phases with a fixed probability λ . The PTs do not generate new data until they transmit the current one. All PDUs have the same size and are transmitted within a single time slot.

C. Mathematical Model of Random Access Phase

We denote the number of unsuccessful RA attempts for the d -th PT as b_d , $d = 1, \dots, D$, and identify the system state at the beginning of the i -th RA as a vector $\mathbf{Z}(i) = (Z_0(i), Z_1(i), \dots, Z_{b_{\max}}(i))$, where $Z_0(i)$ is the number of PTs with no data to transmit; $Z_1(i)$ is the number of PTs trying to transmit data at the RA for the first time; $Z_2(i)$ denotes the number of devices that trying to transmit data at the RA for the second time and have $b_d = 1$. The similar applies for $Z_b(i)$, $b = 3, \dots, b_{\max} - 1$. The number of devices experiencing $b_{\max}$ or greater RA transmission attempts is denoted by $Z_{b_{\max}}(i)$. Since every PT is counted in a single component $Z_b(i)$, we have $\sum_{b=0}^{b_{\max}} Z_b(i) = D$. We are also interested in tracking the system state *within* a given RA phase. So, considering some i -th RA we let vector $\mathbf{X}(s) = (X_0(s), X_1(s), X_2(s), \dots, X_{b_{\max}}(s))$ define the system state at each slot $s = 1, \dots, N$ and let $\mathbf{X}(1) = \mathbf{Z}(i)$. The state space

$\mathcal{X}$ is given by

$$\mathcal{X} = \{(X_0, X_1, \dots, X_{b_{\max}}) : X_b \geq 0, \quad (2)$$

$$b = 1, \dots, b_{\max}; \sum_{b=0}^{b_{\max}} Z_b(i) = D\} \quad (3)$$

Next, we show that $\mathbf{X}(s)$ is the inhomogeneous finite Markov chain, and define its evolution over $s = 1, \dots, N$. Then, we show that $\mathbf{Z}(i)$ is the infinite homogeneous Markov chain, and define its transition probabilities.

Assume that d -th PT having $b_d = b$ while passing RA transmits PDU at slot s with probability $q_b(s)$

$$q_b(s) = \begin{cases} 2 [\min(N-1, CW_{\min} 2^b)]^{-1}, & s = 1, \dots, \min(N-1, CW_{\min} 2^b); \\ 0, & s = \min(N-1, CW_{\min} 2^b) + 1, \dots, N, \end{cases} \quad (4)$$

This formulation ensures that: (i) a PT does not attempt transmission beyond its contention window; (ii) the average backoff is approximately half of the contention window size; (iii) a PT may re-attempt transmission within the same RA phase if the channel is busy (due to collision occurrence); and (iv) a PT may also defer its transmission if it senses the channel is busy (due to LBT). We assume successful transmission occurs only when a single PT attempts to transmit in one slot. Therefore, for d -th PT having $b_d = b$ at slot s define the successful transmission probability $p_b(s)$ as

$$p_b(s) = X_b(s) q_b(s) (1 - q_b(s))^{X_b(s)-1} \cdot \prod_{\hat{b} \in \{1, \dots, b_{\max}\} \setminus \{b\}} (1 - q_{\hat{b}}(s))^{X_{\hat{b}}(s)}, \quad (5)$$

$$b = 1, \dots, b_{\max}, \quad s = 1, \dots, N.$$

It follows that system state $\mathbf{X}(s)$ at slot $s = 2, \dots, N-1$ depends only on system state $\mathbf{X}(s-1)$ at slot $s-1$ and the slot number s . For slot $s = 1, \dots, N-2$ the probability of transition from state $\mathbf{X}$ to state $\hat{\mathbf{X}}$ at slot $s+1$ is defined by

$$P_s(\mathbf{X}, \hat{\mathbf{X}}) = \begin{cases} p_1(s), & \mathbf{X} = (X_0, X_1, \dots, X_{b_{\max}}), \\ & \hat{\mathbf{X}} = (X_0 + 1, X_1 - 1, \dots, X_{b_{\max}}); \\ p_2(s), & \mathbf{X} = (X_0, X_1, X_2, \dots, X_{b_{\max}}), \\ & \hat{\mathbf{X}} = (X_0 + 1, X_1, X_2 - 1, \dots, X_{b_{\max}}); \\ \dots & \\ p_{b_{\max}}(s), & \mathbf{X} = (X_0, X_1, \dots, X_{b_{\max}}), \\ & \hat{\mathbf{X}} = (X_0 + 1, X_1, \dots, X_{b_{\max}} - 1); \\ 1 - \sum_{b=1}^{b_{\max}} p_b(s), & \mathbf{X} = \hat{\mathbf{X}}; \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

At the end of the RA phase (after slot $N-1$), any PT that has not successfully transmitted its PDU increments its

backoff stage counter, i.e., $b_d \leftarrow b_d + 1$. Then, the system state at slot N is defined by

$$X_0(N) = X_0(N-1), \quad (7)$$

$$X_1(N) = 0, \quad (8)$$

$$X_{b+1}(N) = X_b(N-1), \quad b = 1, \dots, b_{\max} - 1, \quad (9)$$

$$X_{b_{\max}}(N) = X_{b_{\max}-1}(N-1) + X_{b_{\max}}(N-1). \quad (10)$$

Here, $X_1(N) = 0$ as all devices with a PDU have failed at least once and there has been no new data generated yet. After this step, until the next RA, each PT with no PDU generates data independently with fixed probability λ , so we can derive the probability $A(\mathbf{X}, \hat{\mathbf{X}})$ of the system changing its state from $\mathbf{X}$ at N -th slot to the $\hat{\mathbf{X}}$ at the first slot of the next RA

$$A(\mathbf{X}, \hat{\mathbf{X}}) = \begin{cases} \binom{X_0}{k} (\lambda^k) (1 - \lambda)^{(X_0 - k)}, & k = 0, \dots, X_0, \\ & \hat{\mathbf{X}} = (X_0 - k, k, X_2, \dots, X_{b_{\max}-1}, X_{b_{\max}}), \\ & \mathbf{X} = (X_0, 0, X_2, \dots, X_{b_{\max}-1}, X_{b_{\max}}); \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

Ordering all feasible states $\mathcal{X}$ we compose $|\mathcal{X}| \times |\mathcal{X}|$ matrixes $\mathbf{P}_s$ with elements $P_s(\mathbf{X}, \hat{\mathbf{X}})$ for all $s = 1, \dots, N-2$, and matrix $\mathbf{A}$ with elements $A(\mathbf{X}, \hat{\mathbf{X}})$. Also, compose matrix $\mathbf{R}$, where element $R(\mathbf{X}, \hat{\mathbf{X}})$ is equal to 1 if system being in state $\mathbf{X}$ at $(N-1)$ -th slot changes its state to $\hat{\mathbf{X}}$ according to (7)-(10). Define matrix $\mathbf{P}$ by

$$\mathbf{P} = \left(\prod_{s=1}^{N-2} \mathbf{P}_s \right) \mathbf{R} \mathbf{A}. \quad (12)$$

As all RA phases are performed in the same way, $\mathbf{P}$ is the transition probability matrix for Markov chain $\mathbf{Z}(i)$, $i = 1, 2, \dots$

III. HIDDEN MARKOV MODEL

We have specified the Markov chain $\mathbf{X}(s)$, describing RA operation. However, the FT that is responsible for radio resource management can not capture the complete system state in RA slots, as it does not know which PT has a PDU for transmission or which PTs are attempting to transmit. Instead, at each RA slot, it is possible for the FT to measure whether the slot contains any transmission, whether it causes a collision, or whether a successful transmission occurs within that slot. If a successful transmission is made from the d -th PT, the FT can identify the PT's attempt number from

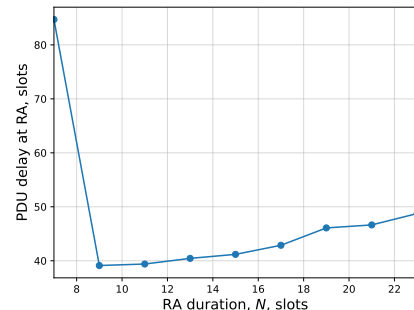


Fig. 2: Mean delay at the RA phase

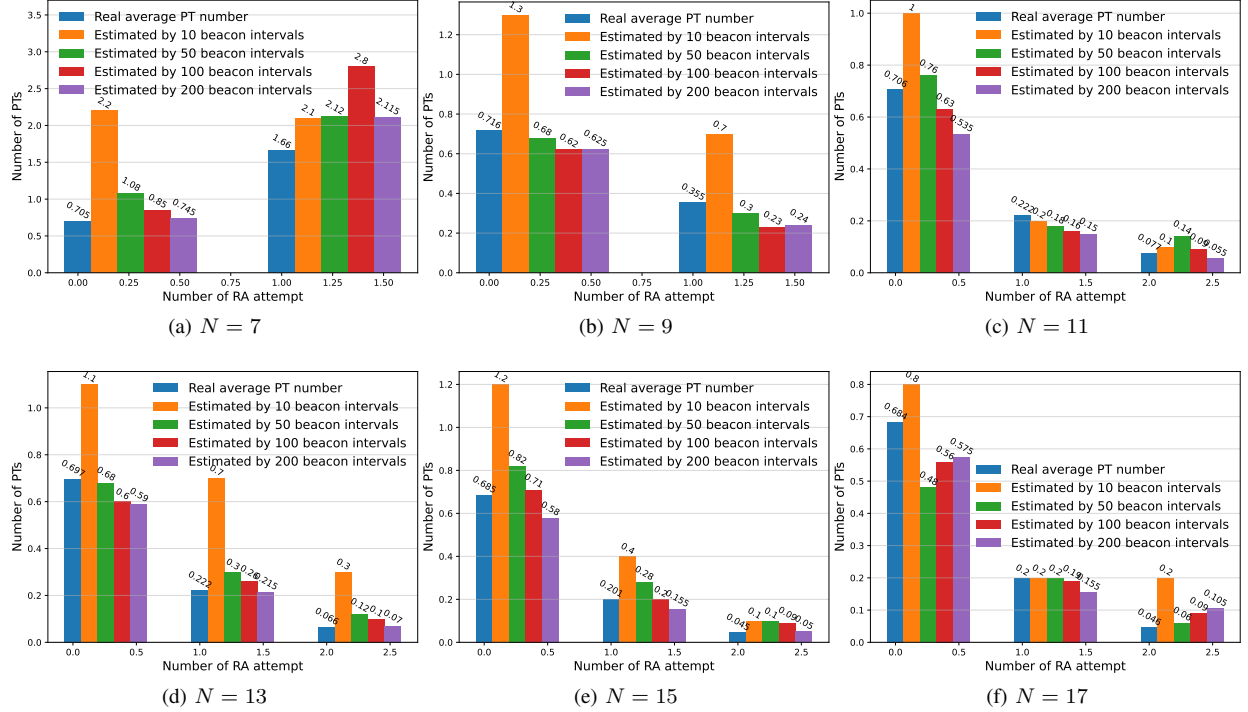


Fig. 3: Estimation of the mean number of PTs finishing different RA attempts at the end of RA

the received PDU. Let the set of possible FT observations be $\mathcal{O} = \{0, 1, \dots, b_{\max} + 1\}$, where 0 denotes empty slot, $b \in \{1, \dots, b_{\max}\}$ denotes attempts number for device that successfully transmits PDU, and $b_{\max} + 1$ denotes collision slot. Let $Y(s)$, $s = 1, \dots, N$ be the FT's observation obtained at slot s . Define $M(s, \mathbf{X}, o)$ as the probability of measuring observation o at RA slot $s = 1, \dots, N - 1$ when the system state is given by $\mathbf{X}(s) = \mathbf{X} = (X_0, X_1, \dots, X_{b_{\max}})$.

$$M(s, \mathbf{X}, 0) = \prod_{\hat{b}=1}^{b_{\max}} (1 - q_{\hat{b}}(s))^{X_{\hat{b}}}, \quad (13)$$

$$M(s, \mathbf{X}, b) = p_b(s), \quad b = 1, \dots, b_{\max}, \quad (14)$$

$$M(s, \mathbf{X}, b_{\max} + 1) = P_s(\mathbf{X}, \mathbf{X}) - M(s, \mathbf{X}, 0), \quad (15)$$

and

$$M(N, \mathbf{X}, 0) = 1. \quad (16)$$

The problem is in identifying $\tilde{\mathbf{X}}$, which denotes the most probable system state at $N - 1$ RA slot. This should be determined with respect to a given sequence of observations $\{Y(s)\}_{s=1}^N = Y_1, Y_2, Y_3, \dots, Y_{N-1}$ that were collected by the FT during every RA slot from $s = 1$ to $N - 1$, given $P_s(\mathbf{X}, \tilde{\mathbf{X}})$, $M(s, \mathbf{X}, o)$, and probabilities $\pi(\mathbf{X})$ for the system to be in state $\mathbf{X}$ at the first slot. In case, when new PDU generation rate λ is known for advance, one can use stationary probabilities of chain $\mathbf{Z}(i)$ for determining $\pi(\mathbf{X})$. Specifically, this probabilities can be combined in vector π and founded by solving the following system

$$\begin{cases} \pi = \pi \mathbf{P} \\ \pi \mathbf{1} = 1. \end{cases} \quad (17)$$

In case when λ is unknown, one may set $\pi(\mathbf{X})$ to the equal values that are summing up to one over all possible $\mathbf{X}$. Further we assume λ is known. For finding $\tilde{\mathbf{X}}$ we utilize the Viterbi algorithm [14], which can also be found in [15, Appendix A] and in Appendix.

Obtaining the value of $\tilde{\mathbf{X}}$ for K beacon intervals, we can average them to find the average number of PTs that have finished their b -th RA transmission attempt by the end of the RA. In the next section we show how this helps optimize network performance.

IV. NUMERICAL RESULTS

Let the duration between RA phases be equal to 48, number of PTs D be $D = 12$, probability of packet generation λ be $\lambda = 0.15$, minimum backoff interval be $CW_{\min} = 4$. By varying RA duration N and running a simulation of the mathematical model $\mathbf{Z}(i)$ over 5000 beacon intervals, we show the average delay for PTs in the RA phase at the Fig.2 and the mean number of devices finishing their $b = 1, \dots, b_{\max}$ -th RA attempt at the end of the RA in Fig. 3. Note that a PT attempting transmission remains in the RA phase for its entire duration of N slots. For each unsuccessful attempt, it must additionally wait for the subsequent RA phase to begin, accumulating further delay.

We can see from Fig. 2 that, in case $N = 8$, PTs experience large delays during RA passing. This is due to collisions. As we can see in Fig. 3a, more than half of the PTs pass the RA on their second or later attempts. On the other hand, when the number of PTs passing the RA in the b -th attempt exceeds the number passing in the $(b + 1)$ -th attempt, the average delay is greatly reduced. Under these conditions, slight delays increase

with increasing N due to the increased delays of PTs that successfully passed the RA on the first attempt.

In Fig. 3, along with the mean number of PT obtained by simulation, estimates of these values are presented. These were computed by running a simulation of Markov chain $\mathbf{Z}(i)$ for $K \in \{10, 50, 100, 200\}$ beacon intervals, collecting K observations sequences $Y_1, Y_2, \dots, Y_{N-1}$ (one sequence per RA), calculating estimates $\hat{X}$ for K times and averaging them. As expected, increasing the number of observed beacon intervals K leads to more accurate estimates. For instance, with $K = 200$ and $K = 100$, the average estimation error over all cases is less than 25%. Meanwhile a mean error among PT's groups with fixed N is less than 38%.

V. CONCLUSION

The research has addressed the problem of optimizing RA performance in DECT-2020 NR networks. A comprehensive mathematical model describing the RA phase operation was presented and HMM method for estimating PTs waiting times by experienced RA attempts was presented. The results demonstrate that the proposed HMM-based estimation method provides quite accurate predictions of device behavior during the RA phase even by observing 100-200 RA phases. The delay reduction strategy shows significant potential for improving network performance by optimizing resource allocation. The findings presented in this work contribute to the advancement of DECT-2020 technology and provide valuable insights for optimizing mMTC networks.

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APPENDIX

VITERBI ALGORITHM

Algorithm 1 Viterbi Algorithm for State Estimation

Require:

Observation sequence $Y = (Y_1, \dots, Y_{N-1})$;
Transition probabilities $P_s(\hat{\mathbf{X}}, \mathbf{X})$, $s = 2, \dots, N - 1$;
Observation probabilities $M(s, \mathbf{X}, o)$;
Initial state distribution $\pi = (\pi(\mathbf{X}))_{\mathbf{X} \in \mathcal{X}}$.

Return: The most likely final state $\hat{\mathbf{X}}$.

- 1: Let $\text{ind}(\mathbf{X})$ be the index for state $\mathbf{X} \in \mathcal{X}$.
 - 2: Initialize matrices $viterbi[|\mathcal{X}|, N - 1]$ and $backpointer[|\mathcal{X}|, N - 1]$.
 1. **Initialization Step**
 - 3: **for** each state $\mathbf{X} \in \mathcal{X}$ **do**
 - 4: $viterbi[\text{ind}(\mathbf{X}), 1] \leftarrow \pi(\mathbf{X}) \cdot M(1, \mathbf{X}, Y_1)$
 - 5: $backpointer[\text{ind}(\mathbf{X}), 1] \leftarrow 0$
 - 6: **end for**
 2. **Recursion Step**
 - 7: **for** $s = 2$ to $N - 1$ **do**
 - 8: **for** each state $\mathbf{X} \in \mathcal{X}$ **do**
 - 9: $viterbi[\text{ind}(\mathbf{X}), s] \leftarrow$
 $\max_{\hat{\mathbf{X}} \in \mathcal{X}} \left(viterbi[\text{ind}(\hat{\mathbf{X}}), s - 1] \cdot P_s(\hat{\mathbf{X}}, \mathbf{X}) \right) \cdot M(s, \mathbf{X}, Y_s)$
 - 10: $backpointer[\text{ind}(\mathbf{X}), s] \leftarrow$
 $\arg \max_{\hat{\mathbf{X}} \in \mathcal{X}} \left(viterbi[\text{ind}(\hat{\mathbf{X}}), s - 1] \cdot P_s(\hat{\mathbf{X}}, \mathbf{X}) \right)$
 - 11: **end for**
 - 12: **end for**
 3. **Termination Step**
 - 13: $\hat{\mathbf{X}} \leftarrow \arg \max_{\mathbf{X} \in \mathcal{X}} (viterbi[\text{ind}(\mathbf{X}), N - 1])$
 - 14: **return** $\hat{\mathbf{X}}$
-